



NUBRED

AI Contract Extraction Guide

What the AI must identify, extract and structure from every uploaded contract

FOR: Developer (Ali Kumail) · AI Prompt Engineering

6 contract types · 5 extraction layers · Built from 46 real contracts

This document defines the AI extraction schema for the NuBred contract analysis engine.

How to Use This Document

This guide defines exactly what the NuBred AI must extract when a contract is uploaded to the platform. It is structured in two parts:

- Part 1 — Universal Extraction Layer: fields that must be extracted from every contract, regardless of type. These form the Contract Identity Record.
- Part 2 — Contract-Type Extraction Layer: fields specific to each contract family. These populate the Obligation Map for each party.

The goal is that after analysis, any NuBred user — Admin, IVM, or Licensee — can answer these five questions without reading the contract:

1	Who are the parties and what is each party's role?
2	What is each party authorised to do, obligated to do, and prohibited from doing?
3	What are the key dates and deadlines?
4	What does each party pay or receive, and how is it calculated?
5	What happens if something goes wrong — termination triggers, penalties, surviving obligations?

The AI output for each contract must be a structured JSON object. The schema is defined in this document and implemented by the developer. The JSON is stored in the Analysis and ClauseResult tables in the NuBred database.

PART 1 — Universal Extraction Layer

Extract these fields from every contract regardless of type. They form the Contract Identity Record — the top-level object that describes what the contract is before any clause analysis begins.

1.1 Contract Identity		
Field	What to extract / How to extract it	AI output format / Example
Contract type	Identify the family from the 24-family taxonomy. Look at the WHEREAS clauses and the object article. What economic function does this contract perform? Who collects royalties? Who bears commercial risk?	<i>F6 — IVM Development Agreement (Evaluation + Option)</i>
Economic function	Assign one of the 8 macro-functions: MF-1 Propagation / MF-2 Evaluation / MF-3 VMC Development / MF-4 Fruit Distribution / MF-5 Terminal Licence / MF-6 Territory Positioning / MF-7 Germplasm / MF-8 Operational Support	<i>MF-3 — Delegated commercial development</i>
Contract language	Detect the document language. If bilingual (Italian/English), note both. Extract from the bilingual version the English text.	<i>Bilingual: Italian + English</i>
Governing law	Find the Governing Law and Jurisdiction clause. Extract the applicable law and the competent court or arbitration body.	<i>Italian law — Court of Bologna exclusive jurisdiction</i>
Effective date	Find the Effective Date or date of signature. Extract in ISO 8601 format.	<i>2022-07-18</i>

1.2 Parties		
Field	What to extract / How to extract it	AI output format / Example
Party 1 — name	Extract the full legal name of Party 1 (typically the Licensor, Principal, or Breeder). Include legal form (soc. cons. a r.l., Ltd, etc.).	<i>C.I.V. Consorzio Italiano Vivaisti soc. cons. a r.l.</i>
Party 1 — role	Assign the NuBred role from the contract text: Breeder / Principal / Licensor / IVM / Licensee / Nursery / Consultant / LPM / Contractor. Use the NuBred Glossary terms.	<i>Breeder / Principal / Licensor</i>
Party 1 — country	Extract the country of registration from the registered office address.	<i>Italy</i>
Party 2 — name	Extract the full legal name of Party 2.	<i>Agropro Ltd.</i>
Party 2 — role	Assign the NuBred role.	<i>IVM / Contractor</i>
Party 2 — country	Extract the country of registration.	<i>Israel</i>
Additional parties	If more than 2 parties exist (e.g. AoPs in Fruit Supply Agreements, Joint Breeding with 2 institutes), list them with name and role.	<i>Ovocinárske družstvo BONUM — Association of Producers (Slovakia)</i>

1.3 Varieties and IP		
Field	What to extract / How to extract it	AI output format / Example
Variety name(s)	Extract all variety names mentioned. Include commercial name and/or code. If listed in an Annex, note: 'See Annex N — [N] varieties listed'.	<i>Smeralda (CPVO 44847); Murano; Sibilla</i>
PBR status	For each variety, extract: Protected (grant number + office) / Application pending (filing date) / Not yet filed. Source: typically Annex 1 or 2.	<i>Smeralda: CPVO 44847, granted 17/10/2016. Sibilla: CPVO application pending.</i>

Trademark	Extract any trademark(s) licensed. Include TM name, registration number, territory of registration. Note if TM licence is included in the contract or separate.	<i>TM 'Dori' — EUIPO registration pending. Included in this contract.</i>
EDV clause present	Check if the contract contains a clause on Essentially Derived Varieties. Yes/No. If yes, summarise ownership rule in one sentence.	<i>Yes — EDVs discovered by Contractor belong exclusively to CIV. Immediate notification and tissue sample submission required.</i>

1.4 Territories		
Field	What to extract / How to extract it	AI output format / Example
Production territory	Extract the geographic territory where propagation or production activities are authorised. Exact country/region as stated in the contract or Annex.	<i>Israel</i>
Commercialisation territory	Extract the geographic territory where fruit or plants may be commercialised.	<i>European Union + Middle East</i>
Exclusivity regime	For each right granted, state whether it is exclusive or non-exclusive. If exclusivity is conditional (e.g. subject to Minimum Quantities), note the condition.	<i>Propagation: non-exclusive. Commercialisation: exclusive, subject to Minimum Quantities.</i>

1.5 Term and Duration		
Field	What to extract / How to extract it	AI output format / Example
Contract start date	Extract the Effective Date or the date from which the contract obligations begin.	<i>2022-07-18</i>
Contract end date	Extract the expiry date if fixed (e.g. 31/12/2056, end of PBR, end of planting life). If indefinite or linked to PBR, state the trigger.	<i>Until expiry of CPVO PBR in EU territory (expected 31/12/2056)</i>
Evaluation period	If present, extract the duration of the Evaluation / Trial period and its start trigger (e.g. Release Date, quarantine completion).	<i>4 growing seasons from Release Date</i>
Option exercise window	If an Option is present, extract the window during which it can be exercised and the consequence of non-exercise.	<i>Within 60 days of Evaluation Period end. Non-exercise = Principal free to negotiate with third parties.</i>
Renewal mechanism	State whether automatic renewal applies, the notice period required to prevent renewal, and who has the right to not renew.	<i>Automatic annual renewal. 90 days written notice by either party to prevent renewal.</i>

PART 2 — Contract-Type Extraction Layer

The fields below are specific to each contract family. The AI must first classify the contract (Part 1 field 1.1), then apply the corresponding extraction schema from this section. Each section also contains the Obligation Map — a structured view of what each party must do, must not do, and receives.

A IVM / Evaluation Agreement + Option

Contract families: F6, F24

Agreement between the Breeder/NuBred and an IVM for VCU trial management, variety selection, and commercial development mandate in a territory. The IVM bears trial costs in exchange for Option rights and future royalty share.

Obligation Map — who does what in this contract:

Actor	MUST DO	MUST NOT DO	RECEIVES
Breeder / NuBred	Supply propagating material. Provide technical info on NPVs. File and maintain PBRs. Notify IVM of competing proposals (ROFR). Grant Exclusive Negotiation Period after IVM selects NPVs.	Negotiate with third parties in territory during Evaluation Period. Modify Minimum Licensing Conditions unilaterally after Option is granted.	Entry fee (if any). Future royalty share (25% net of withholding tax). Reimbursement of filing costs from royalty stream.
IVM	Manage importation and quarantine (Phase 1). Conduct VCU trials (Phase 2). Report annually: importation, mother plants, propagation, VCU results. Allow Breeder/Technical Manager access to all trial sites. Notify all EDVs/mutations immediately. Pay fees per Annex.	Sell or distribute plants or fruit during Evaluation Period. Sub-contract without written Breeder approval. Make crosses or create hybrids with NPVs. Disclose variety data publicly without written consent.	Right of Exclusive Negotiation (typically 6 months). Option Right on selected NPVs with pre-agreed Minimum Licensing Conditions. Cost reimbursement if Breeder licences non-selected NPVs to third parties.

A.1 Phase Structure

Field	What to extract / How to extract it	AI output format / Example
Phase 1 scope	Extract the description of Phase 1 (importation and quarantine). Who manages it? Who bears the cost? What is the trigger for Phase 1 end (Release Date)?	IVM manages NPV importation + LPS quarantine at own cost. Phase 1 ends at Release Date notified by IVM to Breeder.
Phase 2 scope	Extract the description of Phase 2 (VCU trials). Who executes trials? Duration (growing seasons)? Any minimum trial conditions per Annex?	IVM + Authorised Sub-Contractors. 4 growing seasons. Minimum trial conditions per Annex 7.
Phase 3 trigger	Extract what triggers Phase 3 (commercial licence). Is it the Option exercise? The Exclusive Negotiation? What is the deadline to reach a licence agreement?	Phase 3 triggered by IVM exercising Option or Right of Exclusive Negotiation. 60 days to finalise licence. Failure = Breeder free to operate.
Release Date	Extract the Release Date definition and the trigger event. Is it fixed or event-triggered?	Release Date = date IVM notifies Breeder that quarantine is complete and certified mother plants are constituted.

A.2 Option and Preferential Rights

Field	What to extract / How to extract it	AI output format / Example
Option varieties	Extract which NPVs carry an Option Right (typically in Annex 5). List by name or code. Distinguish from NPVs that only carry Right of Exclusive Negotiation.	Optioned NPVs: Annex 5 (3 varieties). Selected NPVs (Right of Exclusive Negotiation only): Annex 3.

Minimum Licensing Conditions	Extract the pre-agreed MLC terms from Annex 10 or equivalent. At minimum: royalty rate, exclusivity, territory, minimum quantities.	<i>Royalty: 60% Breeder / 40% IVM on net royalties. Exclusivity: exclusive in Production Territory. Min. quantities: Annex 10.</i>
Option exercise window	How long does the IVM have to exercise the Option? From what trigger date? What is the consequence of non-exercise (IVM waives costs)?	<i>60 days from end of Evaluation Period. Non-exercise = irrevocable waiver of cost reimbursement claims.</i>
ROFR on non-selected NPVs	If Breeder licences an NPV not selected by IVM to a third party, does IVM receive cost reimbursement? Extract the specific article and conditions.	<i>Yes — Art. 11.4: Breeder must reimburse IVM importation and PBR costs for non-selected NPVs licensed to third parties.</i>
Exclusive Negotiation Period	Extract the duration and trigger of the Exclusive Negotiation Period. Can the Breeder negotiate with third parties during this period?	<i>6 months from IVM's NPV selection notice. Breeder cannot negotiate with third parties in territory during this period.</i>

A.3 Reporting Obligations

Field	What to extract / How to extract it	AI output format / Example
Report types and deadlines	List all reports the IVM must submit, with content and deadline. Typically: importation report, mother plant report, propagation report, distribution report, VCU report, Annex update.	<i>NPV importation report: annually + on request. Mother plant report: annually by 31 March. VCU report: annually by 31 March. Distribution report: by 31 October.</i>
Sub-contractor approval	Extract the process for approving sub-contractors. Deadline for Breeder response. What happens if Breeder does not respond?	<i>IVM submits sub-contractor details. Breeder has 30 days to endorse. No reply within 30 days = tacit approval.</i>

A.4 Financial Terms

Field	What to extract / How to extract it	AI output format / Example
Fees during trial phase	Are there any fees payable by the IVM during the Evaluation Period? Extract amount, currency, payment trigger. If no fee: state explicitly.	<i>No fee during Evaluation Period (VCU phase). IVM bears all trial costs at own risk.</i>
Cost reimbursement structure	Extract which costs the IVM can recover and when (from royalty stream, from Breeder if third party licence, etc.). Source article.	<i>Importation costs + PBR filing costs deductible from future royalty stream per Art. 6.5 (ARI/CIV model) or reimbursed by Breeder per Art. 11.4 (Agropro model).</i>

B Full Commercial Licence

Contract families: F1, F7, F18

Complete licence agreement granting the Licensee the right to propagate, cultivate and commercialise a protected variety with trademark. The primary long-term commercial contract between Breeder/NuBred and the operating partner.

Obligation Map:

Actor	MUST DO	MUST NOT DO	RECEIVES
Licensor / Breeder	Supply Finished Plants through Authorised Nursery. Maintain PBRs. File new IP rights. Notify Licensee of PBR transfer (1 month notice). Guarantee plant quality per Annex.	License same variety to competing operators in exclusive territory. Modify Trademark Use Policy without advance notice.	Entry fee. Royalty embedded in plant price or per kg. TM royalties (if club system).
Licensee	Purchase plants only from Authorised Nursery. Plant only on declared georeferenced parcels. Meet Minimum Quantities each season. Submit production and commercialisation reports. Apply TM only per TUP. Supply all fruit to Authorised Distributors.	Maintain Mother Plants. Propagate plants independently. Sell plants to third parties. Create Club Strategy with own TM without Breeder consent. Sell fruit outside Commercialisation Territory.	Exclusive commercialisation right in territory. TM licence. Right to propose extension of Minimum Quantities / Opzioni for expanded exclusivity.

B.1 Rights Bundle

Field	What to extract / How to extract it	AI output format / Example
Propagation right	Extract: granted yes/no, exclusive or non-exclusive, territory. If exercised only through Authorised Nursery (not directly by Licensee), state this explicitly.	<i>Granted. Non-exclusive. Territory: Poland. Must be exercised exclusively through Authorised Nursery (CIV member nurseries). Licensee cannot propagate directly.</i>
Sale and delivery right	Extract: granted yes/no, exclusive or non-exclusive, territory, restriction (Authorised Producers only).	<i>Granted. Exclusive in Cultivation Territory (Poland). Restricted to Authorised Producers only.</i>
Cultivation right	Extract: granted yes/no, exclusive or non-exclusive, territory, restricted to Authorised Producers.	<i>Granted. Exclusive. Poland. Authorised Producers only. Plants must be planted in same season — no nursery-style holding permitted.</i>
Commercialisation right	Extract: granted yes/no, exclusive or non-exclusive, territory, TM included or separate.	<i>Granted. Exclusive per Annex 4/5 Options. EU + selected markets. TM included — use subject to TUP.</i>
TM use right	Is TM use a right or an obligation? Is co-branding permitted? Who approves new packaging? Extract the TUP reference.	<i>Right (not obligation). Co-branding with 'Rajpol/Pola' marks pre-approved. All other co-branding requires prior written Licensor consent. TUP: separate DUM document.</i>

B.2 Production Plan and Quantities

Field	What to extract / How to extract it	AI output format / Example
Minimum Quantities	Extract the Minimum Quantities per season/year. Source: Annex 5 or equivalent Production Plan.	<i>Year 1: 10,000 plants. Year 2: 25,000 plants. Year 3+: 50,000 plants. Tolerance:</i>

	Include tolerance (typically -15%) and recovery period (typically 2 years).	-15% per year. Recovery: 2 consecutive seasons.
Minimum Quantities enforcement	What happens if Minimum Quantities are not met? Extract: loss of exclusivity, lump-sum indemnity, termination right. Source article and Annex.	Failure to meet MQ triggers: (1) lump-sum indemnity per Annex 6, AND (2) Licensor right to revoke exclusivity or terminate. 2-year recovery window before enforcement.
Maximum Quantities	If Maximum Quantities are defined, extract them. What happens if Licensee exceeds them?	Max quantities per Annex 5 Options. Exceeding without authorisation = breach.
Options structure	If Options are present (expand exclusivity, change territory, modify MQ), list each Option with its trigger, payment, and consequence.	Option 1: global exclusivity — €200,000 additional entry fee. Option 2: Poland-only exclusivity — no additional fee. Option 3: automatic — default if no election made.

B.3 Financial Terms

Field	What to extract / How to extract it	AI output format / Example
Entry fee	Extract amount, currency, payment date, and refundability conditions. Is it offset against future royalties or separate?	€200,000 — payable within 30 days of signing. Partially refundable if CPVO PBR rejected after filing. Not offset against royalties.
Royalty structure	How is royalty calculated? Per plant / per kg / per hectare / flat annual / embedded in plant price? Extract rate and basis. State whether royalty is included in plant price or invoiced separately.	Royalty embedded in Finished Plant price. €28 per 1,000 plants (Koffler model). OR flat €1,500/ha planted annually (Čirić model).
Reporting and payment calendar	Extract all financial reporting deadlines: commercialisation report date, royalty payment date, penalisation for late payment.	Commercialisation report: 15 January each year. Royalty payment: 28 February. Late payment: 5% annual interest from due date.

B.4 Authorised Nursery and Plant Supply

Field	What to extract / How to extract it	AI output format / Example
Authorised Nursery	Extract the name(s) of the Authorised Nursery/Nurseries from Annex 4 or equivalent. Can the Licensee change nursery and under what conditions?	Authorised Nurseries: CIV member nurseries (Annex 4). Change requires Licensor written consent.
Plant certification requirement	Must plants come from certified propagating material? What certification standard applies? What is the exception for CIV member nurseries?	Certified material required. Exception: if Authorised Nursery is a CIV member, Licensee is not obligated to monitor certification compliance.
Prohibition on unsold plants	What must happen to unsold Finished Plants at end of season? Extract the disposal obligation and any exception.	Licensee must destroy all unsold Finished Plants at end of season. Exception: Licensor may grant written derogation to carry over limited quantities.

c Nursery Propagation Licence

Contract families: F2, F3, F8

Agreement between the Breeder and a nursery authorising propagation of protected varieties. The nursery receives plant material (mother plants or propagating material) and returns Finished Plants. May be annual (F2) or multi-annual with Mother Plant maintenance (F3), or a tolling/contract manufacturing structure (F8).

Obligation Map:

Actor	MUST DO	MUST NOT DO	RECEIVES
Breeder / Principal	Supply Mother Plants or propagating material. Set and communicate annual production plan. Provide technical information. Inspect nursery facilities. Collect census declarations.	Authorise another nursery in same territory to propagate same variety without notice to this nursery.	Royalty per plant (embedded in plant price or separate). Production plan compliance confirmation. Reports.
Nursery / Licensee	Propagate per Production Plan (within min/max). Maintain Mother Plants per authorised ratio. Submit census reports (typically 3-4 per year). Use only authorised propagating material. Destroy unsold or unauthorised plants. Allow Breeder inspection.	Maintain Mother Plants beyond authorised season. Supply plants to unauthorised parties. Exceed maximum quantities without written authorisation. Sub-contract without Breeder consent.	Non-exclusive propagation right. Access to protected variety material. Revenue from plant sales.

C.1 Production Plan

Field	What to extract / How to extract it	AI output format / Example
Minimum order / minimum quantity	Extract the Minimum Order of Propagating Material per season. What is the consequence of failing minimum order?	Minimum order: 50,000 mother plant cuttings per season. Failure triggers Breeder right to supply other nurseries in territory.
Production ratio (Mother Plants to Finished Plants)	If specified, extract the Mother Plant to Finished Plant ratio. Source: Annex or specific article. This ratio is used for compliance verification.	45 Finished Plants per Mother Plant (Aprica variety). 50 Finished Plants per Mother Plant (Sibilla variety).
Census report schedule	Extract all census report deadlines and content requirements. Typically: Mother Plant census, propagation census, sales census, destruction census.	30 Sept (n-2): Mother Plant census. 15 Sept (n-1): Propagation census. 15 Jan (n): Sales census + destruction declaration.
Penalty for undeclared propagation	Extract the specific penalty for producing plants in excess of the declared census or supplying unauthorised parties.	€3/kg for fruit produced from unauthorised plants. 10x declared royalties for false declarations.

C.2 Contract Manufacturing (Conto Lavorazione) — F8 only

Field	What to extract / How to extract it	AI output format / Example
Material ownership	Extract the clause confirming that propagating material (budwood, marze) supplied by the Principal remains the property of the Principal at all times.	All material supplied by CIV remains CIV property throughout. Nursery acts as service provider, not licensee.
Conforming Finished Plants	Extract the quality criteria for Finished Plants. What standard must be met for payment to be due? Extract quantitative criteria.	BIO standard ≥30% — minimum grade 7/10. Payment triggered only for Conforming Finished Plants.

Non-conformity penalty	What happens if the nursery delivers non-conforming plants? Extract the price reduction formula and makeup obligation.	<i>80% price discount on non-conforming plants. Makeup production obligation in following season.</i>
Insurance obligation	Is the nursery required to carry specific insurance (e.g. hail insurance)? Extract coverage type and who bears the premium.	<i>Mandatory hail insurance — nursery bears premium cost.</i>

D Fruit Producer Sublicence

Contract families: F15, F20

Agreement between the IVM/Licensee and a commercial fruit grower. The grower purchases plants from the Authorised Nursery, cultivates the variety on declared georeferenced parcels, and supplies the entire harvest to the Licensee or authorised distributor. This is the downstream commercial contract closest to actual production.

Obligation Map:

Actor	MUST DO	MUST NOT DO	RECEIVES
IVM / Licensee	Select and approve growers. Provide quality standards (TUP / Annex 3). Process fruit supply and pay grower. Notify grower of authorised distributors. Guarantee continuity if TM licence terminated.	Approve growers outside the licensed territory. Modify authorised distributor list without advance notice to growers.	Entire harvest from grower. Control over channel and brand. Royalty collection on behalf of Breeder.
Grower / Licensed Fruit Supplier	Purchase plants exclusively from Authorised Nursery. Plant only on georeferenced Land (Annex 2). Cultivate per quality standards. Supply entire harvest to Licensee / authorised distributor only. Notify Licensee of any PBR/TM infringement observed. Allow inspection of land, plants, accounts.	Reproduce or multiply plants or plant parts. Give plants to third parties. Use TM. Sell fruit to anyone other than Licensee / authorised distributor. Change land/parcel without updating Annex 2.	Revenue from fruit supply (price per season). Protection from uprooting if Licensor changes.

D.1 Orchard Identification

Field	What to extract / How to extract it	AI output format / Example
Georeferenced parcels	Extract all parcels declared in Annex 2 (or equivalent). For each: GPS coordinates / cadastral reference, area (sqm or ha), training system, planting year, irrigation type.	Sheet 20 Parcel 1060 — 16,000 sqm — espalier — 2022 — drip irrigation. [Catanzani, Innovation Fruit Addendum]
New parcel procedure	What is the process for adding a new parcel? Does it require a formal Addendum? Is Licensor approval needed?	New parcels require signed Addendum approved by Licensee and acknowledged by CIV. Annex 1 and 2 updated accordingly.
Land transfer clause	What happens if the grower sells or leases their land? Must the new owner sign a new Fruit Supply Agreement?	Grower must notify Licensee in writing. New owner must sign a new Fruit Supply Agreement — Licensee approval required.

D.2 Exclusive Supply Obligation

Field	What to extract / How to extract it	AI output format / Example
Supply exclusivity	Must the grower supply 100% of harvest to the Licensee / authorised distributors? Are there exceptions for low-quality / industrial processing fruit?	100% of harvest must be supplied to Licensee or authorised Fruit Marketers. Exception: low-quality fruit destined for industrial processing may be sold to any entity.
Authorised distributors	Extract the list of authorised distributors / Fruit Marketers (Annex 3 or equivalent). Can the grower change distributor and under what conditions?	Authorised Marketers: Granfrutta Zani, F.lli Clementi, Kiwi Uno, Minguzzi (Annex 3). Change permitted — notification to Licensee required by 15 June.

Association of Producers clause	If the grower is part of a cooperative/AoPs, must the AoPs sign the agreement? Extract the co-signatory requirement.	<i>If grower is AoPs member, AoPs must co-sign agreement and become a contractual Party assuming all grower obligations.</i>
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D.3 Penalties and Termination

Field	What to extract / How to extract it	AI output format / Example
Penalty per plant	Extract the specific penalty for breach of key obligations (buy plants from unauthorised source, supply fruit to third party, reproduce plants). Per-unit penalty.	<i>€5.00 per plant purchased in breach. Payable jointly by grower and Licensee to CIV within 90 days of notification. Solve et repete applies.</i>
Termination triggers	List all events that give the Licensee (or Breeder/CIV) the right to terminate immediately. Source article.	<i>Art. 7.1: Immediate termination right for any breach of Art. 3.1, 3.2 (a-h), 3.5, 4.1.</i>
Post-termination obligations	What must the grower do after termination? Destroy plants under supervision? Allow Breeder to enter land? Timeline?	<i>Immediate cessation of use of NPV and TM. Plants must be destroyed under CIV supervision. CIV has right to enter land and destroy plants if grower fails to comply.</i>
TM licence termination continuity	If the Licensor's TM Licence Agreement is terminated by the Breeder, what happens to the grower? Extract the continuity protection clause.	<i>CIV commits not to request uprooting. Grower continues supplying to new Licensor appointed by CIV. If no new Licensor found, grower may market fruit with variety denomination but without TM.</i>

E Co-Breeding / Joint Breeding Agreement

Contract families: F19, F23

Agreement between two breeding entities to jointly develop new varieties through crossing programmes. Governs IP ownership of new genotypes, VCU responsibilities, royalty sharing, and Club Project commercialisation. Highest IP complexity of all contract types.

Obligation Map:

Actor	MUST DO	MUST NOT DO	RECEIVES
Party A (e.g. CIV)	Supply Genotypes A for crossing. Bear importation and filing costs in CIV territory. Annual commercial report by 31 July. Share royalties (50% default). Notify Party B of first commercialisation date.	Make Crosses with Party B genotypes outside the Programme without consent. Claim ownership of Party B's Background. Commercialise Party B genotypes in CIV territory without joint agreement.	50% of net royalties on New Genotypes (default). Full retention of Single Club Project TM royalties in CIV territory. All EDVs on CIV Genotypes.
Party B (e.g. ARI)	Execute Crosses per Appendix A. Conduct VCU assessment in Territory. Submit annual Breeding Report by 31 July. Send Genotypes of Particular Interest to Party A. Allow Party A access to breeding facilities (72h notice).	Distribute New Genotype plants to third parties without consent. Make Crosses outside Authorised Crosses list. Claim EDVs on Party A Genotypes.	50% of net royalties on New Genotypes (default). Right to develop Single Club Project in ARI territory. VCU assessment expertise and access to CIV Genotypes.

E.1 Germplasm and IP Ownership

Field	What to extract / How to extract it	AI output format / Example
Background ownership	Extract the confirmation that each party's pre-existing genotypes (Background) remain exclusively owned by that party. No claim by the other party.	<i>CIV Genotypes: exclusively owned by CIV. ARI Genotypes: exclusively owned by ARI. No licence or transfer of Background is granted by this Agreement.</i>
Foreground ownership	Extract the rule for New Genotype ownership. Default: proportional to royalty sharing (50/50 unless otherwise agreed). Extract any deviation mechanism.	<i>New Genotype ownership = royalty sharing ratio (50% ARI / 50% CIV by default). If one party provides more commercial services, ratio adjustable by written agreement.</i>
EDV ownership	Extract the rule for EDVs discovered during the programme. On Party A genotypes? On Party B genotypes? On New Genotypes?	<i>EDVs on CIV Genotypes: exclusively CIV. EDVs on ARI Genotypes: exclusively ARI. EDVs on New Genotypes: co-owned 50/50 per Art. 4.5.</i>
Sideground	Extract the definition and ownership rule for Sideground (IP created outside the programme during its execution).	<i>Sideground: full ownership by the creating party. Non-exclusive licence option available to the other party at market terms.</i>

E.2 Programme Execution

Field	What to extract / How to extract it	AI output format / Example
Authorised crosses	Are the specific cross combinations defined in the Agreement or Appendix? Can new crosses be added? What is the authorisation process?	<i>Authorised crosses listed in Appendix A. New crosses require bilateral written agreement. All undeclared crosses must be notified to CIV.</i>
Intermediate Selection Phase	Extract the definition of the Intermediate Selection Phase. What happens at this stage? Who makes decisions from this point?	<i>Intermediate Selection Phase: breeding stage as per Appendix A from which New Genotype must be shipped to CIV. From</i>

		<i>this phase, breeding decisions are made jointly.</i>
VCU responsibilities	Who conducts VCU in the Territory? Who conducts VCU outside the Territory? How is VCU data shared?	<i>ARI: VCU in Japan territory. CIV: VCU in CIV Commercialization Territory and in countries CIV deems appropriate. All VCU information shared in writing.</i>
Research managers	Are Research Managers appointed? Extract names and contact details if present. Their authority is communication only — they cannot modify the Agreement.	<i>CIV Research Manager: Saverio Orsucci. ARI Research Manager: Satoshi Goto. No authority to modify Agreement.</i>

E.3 Commercialisation and Royalty Sharing

Field	What to extract / How to extract it	AI output format / Example
Default royalty sharing	Extract the default net royalty sharing ratio on plant sales. What is 'net' defined as? Who deducts what before sharing?	<i>Net Royalties = royalties received netted of taxes and agreed costs. Default split: 50% ARI / 50% CIV. Adjusted if one party provides more commercial services.</i>
IP cost deductions	Which costs can be deducted from royalties before sharing? Extract the agreed deductible cost categories.	<i>Deductible before sharing: filing and maintenance costs for PBR/TM in the commercialising party's territory, importation and quarantine costs, certification costs. All deductions must be documented.</i>
Club Project structure	Is a Joint Club Project contemplated? Who develops the Joint TM? Cost sharing for Joint Club? If no agreement on Joint TM, what is the Single Club Project rule?	<i>Joint Club Project: 50/50 cost sharing, 50/50 TM royalties. Single Club Project: established solely in own territory, all TM royalties retained by establishing party. 60-day window for other party to join.</i>
Reporting calendar	Extract all reporting deadlines: Annual Breeding Report, Commercial Report, royalty payment date.	<i>Annual Breeding Report: 31 July each year. Commercial Report: 31 July. Royalty payment: 30 September each year for preceding 12 months.</i>

F Consulting Agreement

Contract families: F9, F10, F11

Agreement for agronomic, commercial or variety management services. The consultant provides defined services to the Breeder in exchange for a fee (fixed monthly or % of royalties managed). Key risk: non-compete and breeding limitation clauses post-termination.

Obligation Map:

Actor	MUST DO	MUST NOT DO	RECEIVES
Breeder / Principal	Define scope and deliverables. Pay fees per agreed structure and deadlines. Provide access to variety and market information needed to deliver services.	Assign contract to third party without Consultant consent. Modify fee structure unilaterally during contract.	Defined deliverables (variety management reports, trial coordination, billing management, VCU results). IP foreground from any discovery during engagement.
Consultant	Deliver services per scope and Annexes. Maintain strict confidentiality (survives termination). Notify Breeder of any IP infringement discovered. Submit reports per schedule. Obtain Breeder approval before sub-contracting.	Work for Breeder competitors during contract (exclusivity). Breed or develop varieties post-termination using knowledge acquired (breeding limitation). Sub-contract without written Breeder consent. Disclose confidential information to third parties.	Fee (monthly fixed or % of royalties managed per variety). Expense reimbursement (if provided). Right to continue working with named authorised sub-contractors.

F.1 Scope and Deliverables

Field	What to extract / How to extract it	AI output format / Example
Service scope	Extract the defined scope of consulting services. Variety management? Agronomic coordination? Commercial development? Trial management? Billing of nurseries? List all activities.	<i>Variety Management Services: manage nursery billing, coordinate trials, manage licensee relationships, bill vivai directly (first 2 seasons) then via CIV. Additional: trial coordination in Morocco and Huelva.</i>
Deliverables and reports	Extract all deliverables and their delivery dates. What must the consultant produce and when?	<i>Quarterly variety status reports. Annual billing reconciliation. VCU trial summary per variety by 31 March. Updated licensee database quarterly.</i>
Additional services	Are additional services contemplated beyond the base scope? What authorisation process applies? Are additional fees defined?	<i>Additional Services per Art. 3.5: require written agreement + fee update. Trial management outside Territory requires separate written authorisation.</i>

F.2 Fee Structure

Field	What to extract / How to extract it	AI output format / Example
Fee type	Is the fee (a) fixed monthly, (b) % of royalties collected per managed variety, or (c) mixed? Extract the basis clearly.	<i>% of royalties per variety: ~41% on Murano royalties, ~33% on Sibilla royalties. [Idris model] OR Fixed €1,000/month × 10 months. [Arbagri model]</i>
Payment schedule	Extract payment frequency, payment trigger, and invoice procedure. When are fees due?	<i>Annual payment within 30 days of Breeder receiving royalties from managed varieties. Consultant invoices Breeder after receipt of royalty statement.</i>

Expense reimbursement	Are expenses reimbursable? Which categories? Is prior approval required? Is there a cap?	<i>No expense reimbursement per Art. 7.1 (Idris model). All consultant expenses at own cost and included in fee.</i>
Taxation	Who bears local taxes on consultant fees? Is VAT/IVA applicable? Extract the tax clause.	<i>Each party bears own taxes. Consultant responsible for local taxation compliance. Fee stated exclusive of VAT where applicable.</i>

F.3 Non-Compete and Breeding Limitations

Field	What to extract / How to extract it	AI output format / Example
Exclusivity during contract	Is the consultant exclusive to the Breeder? Or can they work for other clients? If exclusive, is there a geographic scope?	<i>Exclusive to CIV for variety management services in Territory during contract. Can work for non-competing clients outside Territory with CIV written consent.</i>
Non-compete post-termination	Extract the duration and scope of the post-termination non-compete. What activities are prohibited? What territory?	<i>1 year post-termination. Prohibition: any variety management or commercialisation activity for CIV competitors in Territory.</i>
Breeding limitations	Is the consultant prohibited from breeding new varieties post-termination using knowledge acquired? Duration? Scope?	<i>Breeding Limitations: post-termination prohibition on breeding NPV-derived varieties or varieties using CIV know-how. Indefinite for unprotected NPVs.</i>
Sub-contracting limitations	Extract sub-contracting rules: which collaborators are pre-authorised, which require approval, change of control clause.	<i>Pre-authorised: El Bi Crop Bulgaria, Hannah Aelterman. Others require CIV written consent. Change of control clause: if consultant is acquired, CIV may terminate immediately (exception: children of consultant).</i>

PART 3 — Survival and Risk Layer

Extract these fields from every contract. They govern what happens at the end and what survives. These are the clauses that matter most to a project manager at contract renewal or in a dispute.

3.1 Termination Triggers

Field	What to extract / How to extract it	AI output format / Example
Express termination clause	Is there an express termination clause (art. 1456 Italian civil code or equivalent)? List all triggering events. These allow immediate termination without prior notice.	<i>Express termination for: failure to meet Minimum Quantities (2 consecutive seasons), unauthorized propagation, non-payment >30 days, material breach of reporting obligations.</i>
Ordinary termination	Under what conditions can either party terminate with notice? What is the notice period?	<i>Either party: 90 days written notice. Breeder: can terminate if PBR lapses globally and no equivalent protection exists.</i>
Termination for Minimum Quantities failure	Specifically: what happens when Minimum Quantities are not met? Is there a grace period? A lump-sum indemnity? A revocation of exclusivity before full termination?	<i>Step 1: lump-sum indemnity per Annex 6. Step 2 (if repeated): revocation of exclusivity. Step 3 (if persisted): termination. 2-year recovery window between Step 1 and Step 2.</i>

3.2 Penalties

Field	What to extract / How to extract it	AI output format / Example
Penalty schedule	Extract all contractual penalties: amount, triggering event, payer, payee. Distinguish between (a) lump-sum indemnity for Minimum Quantities failure and (b) penalty for breach of specific obligations.	<i>Minimum Quantities failure: lump-sum per Annex 6 formula. Unauthorised propagation: €3/kg fruit or 10x declared royalties (Koffler). Plant breach: €5/plant (Boni Bonum). Generic breach: €20,000/ha (Innovation Fruit).</i>
Solve et repete	Does the contract contain a solve et repete clause? For which obligations? What does it mean for the liable party?	<i>Yes — Art. 6.2: penalty payments subject to solve et repete. Party must pay before disputing. No exception permitted.</i>

3.3 Surviving Obligations

Field	What to extract / How to extract it	AI output format / Example
Confidentiality survival	How long does confidentiality survive after contract termination? Is it indefinite or fixed term? Does it cover VCU data, variety data, commercial data?	<i>Confidentiality survives indefinitely post-termination for all Confidential Information. VCU data: survives 20 years post-termination (ARI/CIV co-breeding model).</i>
IP ownership survival	Do IP ownership clauses survive termination? Specifically: EDV ownership, foreground ownership, non-challenge obligation.	<i>IP ownership clauses survive termination indefinitely. Non-challenge obligation survives. EDV notification obligation survives 5 years post-termination.</i>
Plant destruction / return	What must happen to plants, mother plants, and propagating material at contract end? Who supervises destruction? What is the timeline?	<i>All plants must be destroyed under Breeder supervision within 30 days of termination. Mother Plants: immediate destruction. Breeder has right to enter premises to verify.</i>
Non-compete survival	Does any non-compete or breeding limitation survive? Duration post-termination? Geographic scope?	<i>Non-compete: 1 year post-termination. Breeding limitations: indefinite for NPVs accessed during engagement.</i>

3.4 Dispute Resolution

Field	What to extract / How to extract it	AI output format / Example
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Governing law	Extract the applicable law. If bilingual contract with potential conflict between language versions, note which language prevails.	<i>Italian law. In case of discrepancy between Italian and English/Spanish text: Italian text prevails.</i>
Jurisdiction / arbitration	Extract the competent court or arbitration body. Is there any unilateral right to derogate from the agreed forum?	<i>Exclusive jurisdiction: Court of Bologna. Exception: CIV may unilaterally derogate and bring proceedings before any other competent court (Art. 16 Arbagri).</i>
Arbitration	If arbitration is chosen instead of court: extract the arbitration rules, seat, language, number of arbitrators, standard of decision.	<i>ICC Arbitration Rules. Seat: Slovakia. Language: Slovak. 3 arbitrators. Decision in conformity to law (no ex aequo et bono). [Boni Bonum model]</i>

PART 4 — AI Output Format

The AI must return the extracted data as a structured JSON object. Below is the schema. The developer implements this as the response format from the Anthropic API call. All fields are required — if a field is not found in the contract, return null with a note.

JSON key	Type	Description
contract_type	string	Family code: F1-F24
economic_function	string	MF-1 through MF-8
confidence_score	float	0.0-1.0 — classification confidence
governing_law	string	Country of applicable law
effective_date	string	ISO 8601 date
end_date	string null	ISO 8601 date or trigger description
parties	array	Each: {name, role, country}
varieties	array	Each: {name, code, pbr_status, pbr_office, pbr_number}
trademark	object null	{name, registration, territory, included_in_contract}
territories	object	{production, commercialisation, exclusivity_regime}
rights_granted	array	Each: {right_type, exclusive, territory, conditions}
minimum_quantities	object null	{per_season, tolerance_pct, recovery_years, enforcement}
financial_terms	object	{entry_fee, royalty_structure, payment_calendar}
option_rights	object null	{optioned_varieties, mlc_terms, exercise_window, rofr}
reporting_obligations	array	Each: {report_type, deadline, content_summary}
termination_triggers	array	Each: {event, type: express ordinary, notice_days}
penalties	array	Each: {event, amount, unit, payer, payee}
surviving_obligations	array	Each: {obligation, duration, notes}
edv_clause	object null	{present: bool, ownership_rule, notification_required}
dispute_resolution	object	{law, forum, arbitration: bool, arbitration_details}
flags	array	Each: {clause_id, severity: HIGH MEDIUM LOW, type: ABSENCE ANOMALY IMBALANCE, note}

SYSTEM PROMPT INSTRUCTION — include this in every Anthropic API call for contract analysis:

You are a contract analysis engine for NuBred, a variety governance platform. When a contract is uploaded, extract all information according to the NuBred extraction schema. Return a single valid JSON object. Do not include any text outside the JSON. For each field not found in the contract, return null. For the flags array, include all clauses that are absent, anomalous, or imbalanced based on the NuBred Clause Library. Always use NuBred Glossary terminology in all text fields. Never use synonyms not defined in the NuBred Domain Glossary.